// generative instrument

Arc_OS

diceFriend / quick start

Spectral phase distortion, four-operator FM, a pressure resonator and a centred sub interlock across six tuning worlds.

START: USB-C power + 3.5 mm stereo audio out. Begin with the listening level low.

AS: autonomous four-part rhythm. TS: T1 spectral / T2 FM / T3 pressure; T1 + T3 opens sub.

// SIX KNOBS / TWO LAYERS

P	NORMAL	SHIFT
1	Root: MIDI note 24-84	Tempo: 24-240 BPM
2	Arc: phase / spectral / FM	Stretch: spacing / FM ratios
3	Spread: melodic range	Memory: inertia / persistence
4	Contour: focus / envelope	Scale: 6 tuning worlds
5	Coupling: interaction	Glitch: capture / reverse / freeze
6	Decay: material sustain	Exciter: attack / brightness

// TUNING WORLDS

19 EDO subset / 24 EDO subset / 31 EDO subset / 13-step Bohlen-Pierce / septimal unequal / 22 EDO subset. ROOT transposes without pulling intervals toward 12-TET.
SHIFT: up = second layer; down = normal. Pickup: after changing layers, using DICE or receiving CC, move a knob to take control.

// DICE / MODES

Tap DICE: related mutation; root and tempo stay fixed. SHIFT + tap DICE: full mutation except root / tempo. Hold DICE 1.8 s: AS / TS. SHIFT + DICE 10 s: bootloader.

// PADS / CLOCK

AS pads add spectral, FM and pressure accents. TS pads hold those voices; T1 + T3 holds sub. Internal clock advances AS only. External CV or MIDI Clock can rotate through the four TS materials. Priority: CV > MIDI > internal; MIDI returns after 20 s, internal after 30 s.

// CV IN

CV1: priority trigger / clock; accepted periods generate USB Start + 24 PPQN Clock. CV2: smoothed material modulation and CC74. Banana: 0-9 V; common GND.

// USB MIDI

IN: ch 1-4 address spectral / FM / pressure / sub; ch 5-16 allocate free voices. Note On strikes; Note Off releases. OUT: four channels with Pitch Bend + Note On / Off. Panel CC20-31; CV2 CC74.

// FIRST PATCH

Choose 19 EDO or 24 EDO. Keep ARC / COUPLING below one third, GLITCH near zero and MEMORY / DECAY near the centre. Raise STRETCH, then add EXCITER and a little GLITCH.

// LEDS

Layer LEDs show NORMAL / SHIFT. Lower LEDs 1-4 show spectral / FM / pressure / sub activity; DICE and mode changes flash mutation status.